

Overview of Statistical Learning and Modeling – 36-600

Lab 7T: Experimental Design and One-Way ANOVA

Due Wednesday, October 8, 11:59pm

Trishla Nair

Data: Part I

We'll begin by importing the breast cancer dataset we've looked at previously:

```
suppressMessages(library(tidyverse))
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

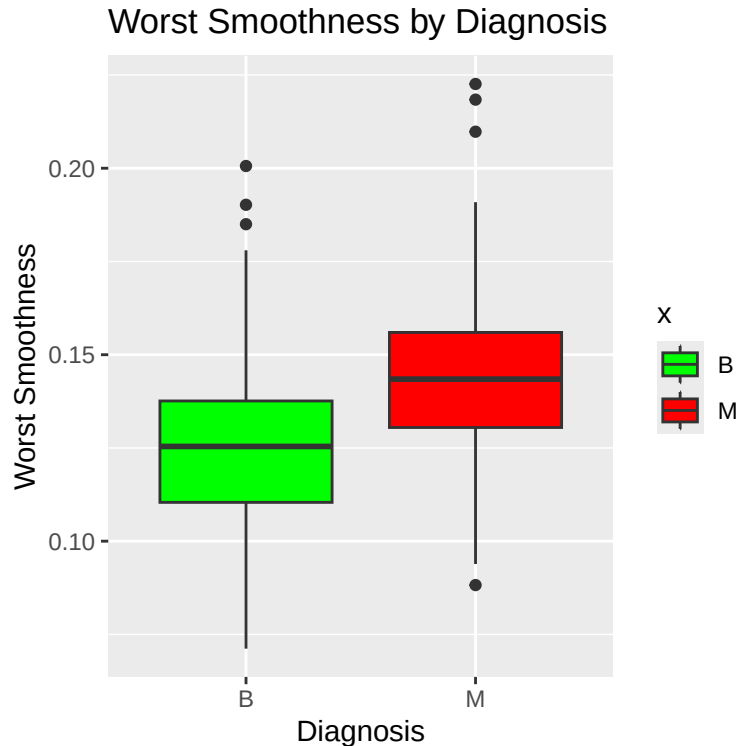
```
df      <- read.csv("https://raw.githubusercontent.com/FSKoerner/F25-36600-data/refs/heads/main/breast_cancer_data.csv")
df      <- df[, c(1, 16)]
names(df) <- c("x", "Y")
```

We retained `Diagnosis` as our categorical predictor variable (now dubbed `x`) and `Worst.Smoothness` as the response variable (now dubbed `Y`).

Question 1

Create a grouped boxplot and make a mental note as to whether or not it appears that the true means for each group are different.

```
ggplot(df, aes(x = x, y = Y, fill = x)) +
  geom_boxplot() +
  labs(title = "Worst Smoothness by Diagnosis",
       x = "Diagnosis",
       y = "Worst Smoothness") +
  scale_fill_manual(values = c("B" = "Green", "M" = "Red"))
```



Question 2

Now let's determine if the assumptions underlying the use of the two-sample t test hold. Specifically, is it reasonable to assume the data within the individual groups are normally distributed with constant variance σ^2 ?

First, run a `shapiro.test()` on the data of groups B and M. If the p -value is less than $\alpha = 0.05$, we reject the null hypothesis that the data are normally distributed. (However, even if we reject the null hypothesis, we will still use these data below for illustrative purposes!)

Recall that in `dplyr` you can do the test as follows:

```
df %>% filter(., x == "<INSERT CATEGORY NAME>") %>% select(., Y) %>% pull(.) %>% shapiro.test(.)
```

The `pull()` function (which is new to us, I think...) coerces a column of numbers from a data frame into a the somewhat-simpler structure of a *vector*, which we need to do, as the Shapiro-Wilk test function doesn't take data frames as inputs.

Go ahead and perform the test here, then state a conclusion.

```
print(df %>% filter(., x == "B") %>% select(., Y) %>% pull(.) %>% shapiro.test())
```

```
##
##  Shapiro-Wilk normality test
##
## data:  .
## W = 0.99046, p-value = 0.02047
```

```
print(df %>% filter(. , x == "M") %>% select(. , Y) %>% pull(.) %>% shapiro.test(.))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: .  
## W = 0.98487, p-value = 0.02298
```

Based on the Shapiro-Wilks test results of the 2 groups B and M, We can say the 2 groups are or very mu

Second, run the `leveneTest()` (from the `car` package; you might have to install this!). Here, you need not separate the data – you can simply input a model formula ($Y \sim x$) and specify `data = df`. If the p -value is less than $\alpha = 0.05$, we reject the null hypothesis that the variances are the same across groups. What do you conclude? (We should note a rule of thumb that you may come across: two-sample t tests and ANOVA are relatively robust to heterogeneity of variance so long as the largest variance is no more than four times the smallest variance. So even if we reject the null here, we might very well still be able to pursue ANOVA-based results.)

```
library("car")
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
##  
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':  
##  
##      recode
```

```
## The following object is masked from 'package:purrr':  
##  
##      some
```

```
leveneTest(Y ~ x, data = df)
```

```
## Levene's Test for Homogeneity of Variance (center = median)  
##      Df F value Pr(>F)  
## group 1    0.532 0.4661  
##      567
```

since the p -value is much larger than the alpha of 0.05, we fail to reject the null hypothesis that the

Question 3

Now, run a two-sample t test and make a conclusion about whether the response values for each group (B and M) have the same mean. Note that like for Levene's test, the `t.test()` will allow you to input a model formula, so you don't need to split up the data yourself. The null hypothesis, by the way, is that the difference in means is equal to zero.

```
t.test(Y ~ x, data = df, var.equal = TRUE)

##
## Two Sample t-test
##
## data: Y by x
## t = -11.067, df = 567, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group B and group M is not equal to 0
## 95 percent confidence interval:
## -0.02341512 -0.01635636
## sample estimates:
## mean in group B mean in group M
## 0.1249595 0.1448452
```

Based on the extremely small p-value we reject the null hypothesis that there is no difference in between

Data: Part II

We'll continue now by importing the hospital cost dataset we've looked at previously:

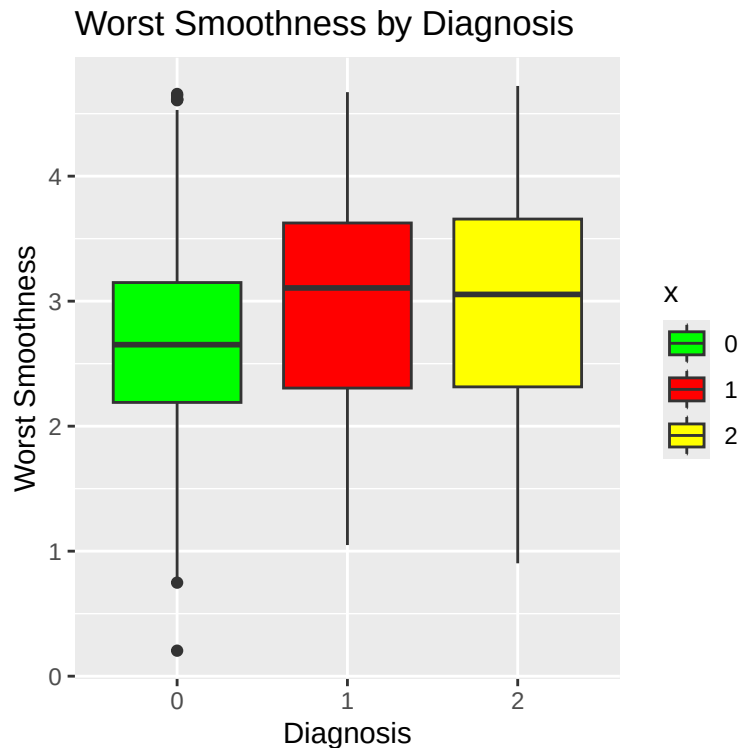
```
df <- read.csv("https://raw.githubusercontent.com/FSKoerner/F25-36600-data/refs/heads/main/hospital.csv")
df <- df[, c(5, 1)]
names(df) <- c("x", "Y")
df$x <- factor(df$x)
w <- which(df$Y == 0)
df <- df[-w, ]
df$Y <- log10(df$Y)
```

We've retained `Drugs` as our categorical predictor variable (now dubbed `x`) and `Cost` as the response variable (now dubbed `Y`). The groups for `x` are "0", "1", and "2". Note that we transformed `Y` logarithmically after removing values of zero - this is one way effectively impose the log transform when the data are not all strictly positive.

Question 4

Repeat Question 1 here: create a grouped boxplot and make a mental note as to whether or not it appears that the true means for each group are different.

```
ggplot(df, aes(x = x, y = Y, fill = x)) +
  geom_boxplot() +
  labs(title = "Worst Smoothness by Diagnosis",
       x = "Diagnosis",
       y = "Worst Smoothness") +
  scale_fill_manual(values = c("0" = "Green", "1" = "Red", "2" = "Yellow"))
```



Question 5

Repeat Question 2 here: run three Shapiro-Wilk tests and one Levene's test, and state which underlying assumptions of the ANOVA model hold here, and which do not. If Levene's test indicates that the variances are truly unequal, compute the variances for each sample and see whether or not the rule-of-thumb given in Question 2 can be applied here. (To compute variances, just use the same "codeflow" as you used for the Shapiro-Wilk test but substitute the `var()` function at the end.)

```
print(df %>% filter(. , x == "0") %>% select(. , Y) %>% pull(.) %>% shapiro.test())
```

```
##
## Shapiro-Wilk normality test
##
## data: .
## W = 0.99549, p-value = 0.07532
```

```
print(df %>% filter(. , x == "1") %>% select(. , Y) %>% pull(.) %>% shapiro.test())
```

```
##
## Shapiro-Wilk normality test
##
## data: .
## W = 0.9735, p-value = 0.06801
```

```
print(df %>% filter(. , x == "2") %>% select(. , Y) %>% pull(.) %>% shapiro.test())
```

```
##
## Shapiro-Wilk normality test
##
## data:  .
## W = 0.9835, p-value = 0.3197
```

given the large p-values for each group, we fail to reject the null hypothesis and that all the groups have the same mean.

Question 6

Show the summary output from regressing the variable Y upon x. What is the estimated mean response for each group?

```
model <- lm(Y ~ x, data = df)
summary(model)
```

```
##
## Call:
## lm(formula = Y ~ x, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.46244 -0.52889 -0.01008  0.52884  1.99096
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.66656    0.03247  82.111  < 2e-16 ***
## x1           0.30586    0.09133   3.349  0.00085 ***
## x2           0.36439    0.09088   4.010  6.66e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8008 on 782 degrees of freedom
## Multiple R-squared:  0.0301, Adjusted R-squared:  0.02762
## F-statistic: 12.14 on 2 and 782 DF,  p-value: 6.45e-06
```

The estimated mean response per group is 0.30586 and 0.36439.

Question 7

Now pass the output from your call to linear regression into the `anova()` function. What is the hypothesis test statistic value and the p-value, and what conclusion do you draw?

```
anova(model)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df Sum Sq Mean Sq F value    Pr(>F)
## x           2  15.56   7.7817  12.136 6.45e-06 ***
## Residuals 782 501.42   0.6412
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Due to the extremely small p-value, we reject the null hypothesis and there is a difference in means.

Question 8

Finally, pass the output from `aov(Y ~ x, data = df)` to the `TukeyHSD()` function and state a conclusion: which group or groups differ from the others? Which do we conclude have the same means?

```
TukeyHSD(aov(Y ~ x, data = df))
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Y ~ x, data = df)
##
## $x
##      diff      lwr      upr    p adj
## 1-0 0.30585828 0.09140064 0.5203159 0.0024467
## 2-0 0.36439321 0.15099065 0.5777958 0.0001967
## 2-1 0.05853493 -0.22413546 0.3412053 0.8778559
```

Based on the `TukeyHSD()`, The groups that differ from each other are 2 and 1, While 0 and 1 and 0 and 2.